MARK SCHEME · TEACHER COPY

10BR · Chapter 5: The Internet and its Uses

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 46

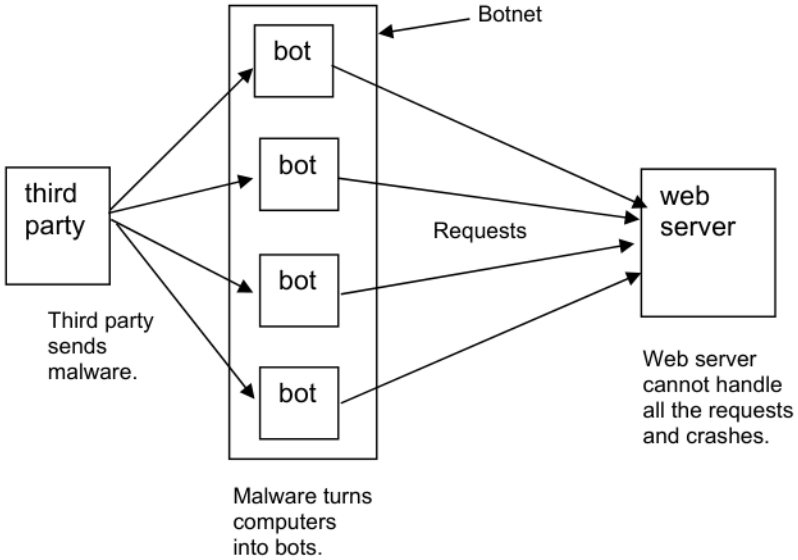
Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Mark scheme extracts © Cambridge University Press & Assessment.

Total: 46 marks

Question 1 · May/June 2024 Paper 12, original Q8 (a),(b) [6 marks]

Question	Answer	Marks
1(a)	**One** mark for each part of the diagram that shows: • A perpetrator/third party sending malware // user downloads/installs malware • Each computer is turned into a bot… • … to create a botnet • Third party initiates the attack • **All** the bots send a request at once to a **web server** • … crashing the webserver Example:  	5
1(b)	Proxy server	1

Question 2 · May/June 2024 Paper 12, original Q11 (a),(b),(c) [8 marks]

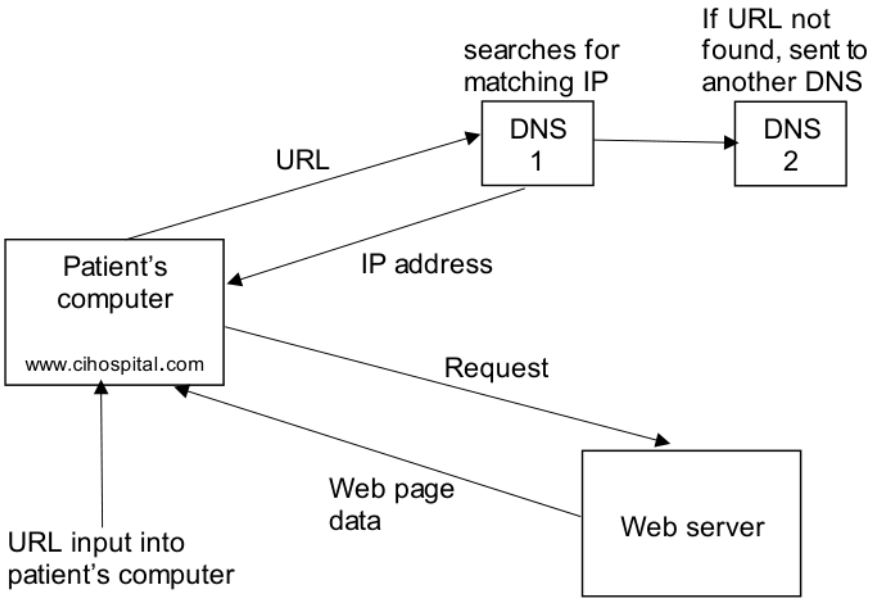
Question	Answer	Marks
2(a)	Any **two** from: • The internet is the infrastructure • The world wide web is a collection of web pages	2

Total: 46 marks

2(b)	Protocol Domain name File **name** // web page **name**	**3**
2(c)	Any **three** from: Example: • Displays a web page • Storing bookmarks/favourites • Recording user history • Allowing use of multiple tabs • Storing cookies • Providing navigation tools • Can set a home page	**3**

Question 3 · May/June 2025 Paper 12, original Q5 (c) [12 marks]

Question	Answer	Marks

3(a)(i)	**One** mark for each correct part of the diagram. The diagram: • Web browser identified as software used to **send** URL/requests or **receives** IP address/web page data • URL/domain sent to DNS • DNS **searches** for **matching** IP address • If not found sent to another DNS • IP address sent from DNS to patient's computer • Request sent from patient's computer to **web server** (for web page) • Web page/HTML data sent from web server to patient's computer  	**6**

3(a)(ii)	Any **six** from: • Encrypted connection established … • … using asymmetric encryption • … to make any data sent meaningless • The **web browser** asks the **web server** to identify itself. • … by sending its digital certificate. • The digital certificate is authenticated/validated by the **web browser**. • If the certificate is authenticated/valid, the connection is secure/the transaction can begin. • If the certificate is not authenticated/invalid, the connection is not secure/the transaction is cancelled/rejected/user is notified	**6**

Question 4 · May/June 2026 Paper 11, original Q6 (a),(b) [10 marks]

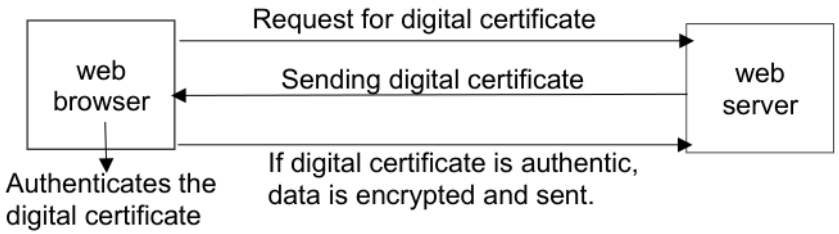
Question	Answer	Marks	Guidance
4	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct point
✗	Incorrect point
?	Unclear response
FT	Follow through
REP	Repetition
I	Ignore
BOD	Benefit of doubt given
NBOD	Benefit of doubt not given
TV	Content of response too vague
NAQ	Not answered question

Annotation	Meaning
λ	Omission
	Section not relevant
	Section incorrect
Highlighter	Information copied from the text
SEEN	Page or response seen by examiner

4(a)	**One** mark for each correct term in the correct place. • domain name server (DNS) • IP address • web browser • request • hypertext markup language(HTML) The user types the URL into the address bar. The web browser sends the domain name from the URL to the **domain name server (DNS)**. This searches for the matching **IP address** and sends this back to the **web browser**. The web browser sends a **request** to the web server for the web page. The web server then sends the data for web page back to the web browser. The web browser renders the **hypertext markup language (HTML)** to display the web page.	**5**	

Total: 46 marks

4(b)	**Five from, one** mark for each correct section of the diagram. • Web browser sending a request to the web server for its (digital) certificate/to identify itself • Web server sending (digital) certificate to the web browser • **Web browser authenticating/validating** the (digital) certificate … • … showing the data sent if it is valid // showing data not sent if it isn't valid • The data is **encrypted // Encryption** is used … • … using **public** and **private** keys // using **asymmetric** encryption • … to make any data sent meaningless • The **web browser** uses the **public key** to **encrypt** the data • The **web server** uses the **private key** to **decrypt** the data For example:  	**5**	MP3 - Checking the certificate/web server is safe is TV. MP3, MP4 and MP5 can be awarded in one annotation e.g. If the web browser thinks the certificate is authentic the encrypted data is sent. MP4 – BOD 'communication begins if certificate is authentic' in place of the data being sent. MP4 – Data could be sent either way, so from browser to server or server to browser MP6 – BOD two **different** keys are used, just 2 keys is TV

Question 5 · May/June 2026 Paper 12, original Q5 (a),(b) [10 marks]

Question	Answer	Marks	Guidance
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		
5(a)	**One** mark for each missing term or description. **term** / **description** brute-force attack: An attack where a third party repeatedly tries to crack a password by entering different combinations. hacking: A person gains **unauthorised/illegal access** to a user's **device/account/system**	**5**	Hacking – stealing/changing data without permission is TV as no reference to gaining access to the device/account. DDoS – BOD just 'denial of service'/'DoS' Spyware – just reference to it tracking the user's activity/actions is TV. Spyware – allow reference to other specific actions than key presses e.g. recording what is captured with webcam or tracking the user's movements from the device's GPS.

	distributed denial of service attack // DDoS	An attack where multiple computers send a large number of requests to a web server with the aim of crashing it.	Data interception - BOD just 'interception' or 'man in the middle'/'MITM'
	spyware	(Malware) that is used to record/track/monitors the keypresses from a user's device.	
	data interception // packet sniffing	A security threat where software is used to analyse data packets during transmission. Any packets that look useful are sent to a malicious third party.	

5(b)	**Five** from: • They **limit/restrict** a user's **access** to data • … by giving users **different permissions** // Users can be given **read only/read-write/no access/execute/administrator** permission • … so an attacker will **only** have access to the data that the **user** has access to • … to stop a **malicious person/attacker deleting/editing/corrupting** data • Prevents a user/attacker from installing malware (as they don't have administrator permissions) • Prevents user/attacker from viewing more **confidential/important** data	**5**	MP1 – BOD controls access to data MP1 – Accept a viable example e.g. a junior employee cannot access customer information MP2 – 'different access levels' is TV for different permissions MP2 – Accept any reference to a named permission. Just low-level or high-level access is TV. MP5 – BOD download in place of install MP6 – Accept an example of confidential data e.g. prevents attacker seeing the company financial accounts. Personal data is TV for an example of confidential data.